Turtle-WoW / Tortoise-WoW Extended - Master Command Reference & Architecture Guide

This document is the canonical CLI operational reference and architecture guide for **Tortoise-WoW Extended** (`twow project/tortoise-wow`), unifying upstream VMaNGOS bugfix porting, native Turtle-WoW core restoration, deterministic bug proving, isolated worktree builds, multi-factor priority ranking, database safety auditing, and release lifecycle management.

> **TIP**
> **Universal Terminal Invocation** All commands can be executed from any terminal, PowerShell console, shortcut, or CI runner without changing current directory: ```powershell & "C:\Users\Admin\AntigravityProfiles\Projects\twow project\tools\task.ps1" <command> [arguments] [options] ``` Or via standard command prompt: ```cmd powershell.exe -ExecutionPolicy Bypass -File "C:\Users\Admin\AntigravityProfiles\Projects\twow project\tools\task.ps1" <command> [arguments] [options] ```

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1. Overview & Operational Principles

The Tortoise-WoW porting system bridges upstream vanilla bugfixes (`vmangos/core`) with the customized Turtle-WoW 1.12.1/1.18.1 server core (`Ildourol/tortoise-wow-extended`).

Fundamental Engineering Invariants:

1. **Deterministic-First Authority**: Heuristics classify; deterministic engines prove; AI assists on ambiguity but NEVER acts as the sole validation gate.

2. **Standardized Exit Codes**:

- `0`: PASS (Operation completed successfully and verified).
- `1`: VALIDATION / COMPATIBILITY FAILURE (Invariant violated, collision, or check failed).
- `2`: TOOL / ENVIRONMENT FAILURE (Missing binary, invalid repo, git error).
- `3`: INTERNAL / SCHEMA FAILURE (Malformed contract, schema validation failure).

3. **Strict Worktree Isolation**: All candidate patches and build checks execute exclusively in dedicated Git worktrees (`.worktrees/PORT-XXXX/`). The user's primary working tree is never touched, reset, or dirtied.

4. **No Destructive Rollbacks**: `git checkout .`, `git reset --hard`, and `git clean -fd` are strictly prohibited on the target repository.

5. **No Production Database Direct Writes**: Audits analyze SQL migrations offline against cached catalog schemas (`config/schema_catalog.json`).

6. **Compile and Link Do Not Imply Startup**: Server health verifies compile, link, and startup/smoke milestones separately.

7. **Single-Writer Constraint**: Compilations through MSVC 2022 / Ninja acquire single-writer synchronization to prevent compiler file locking and git index corruption.

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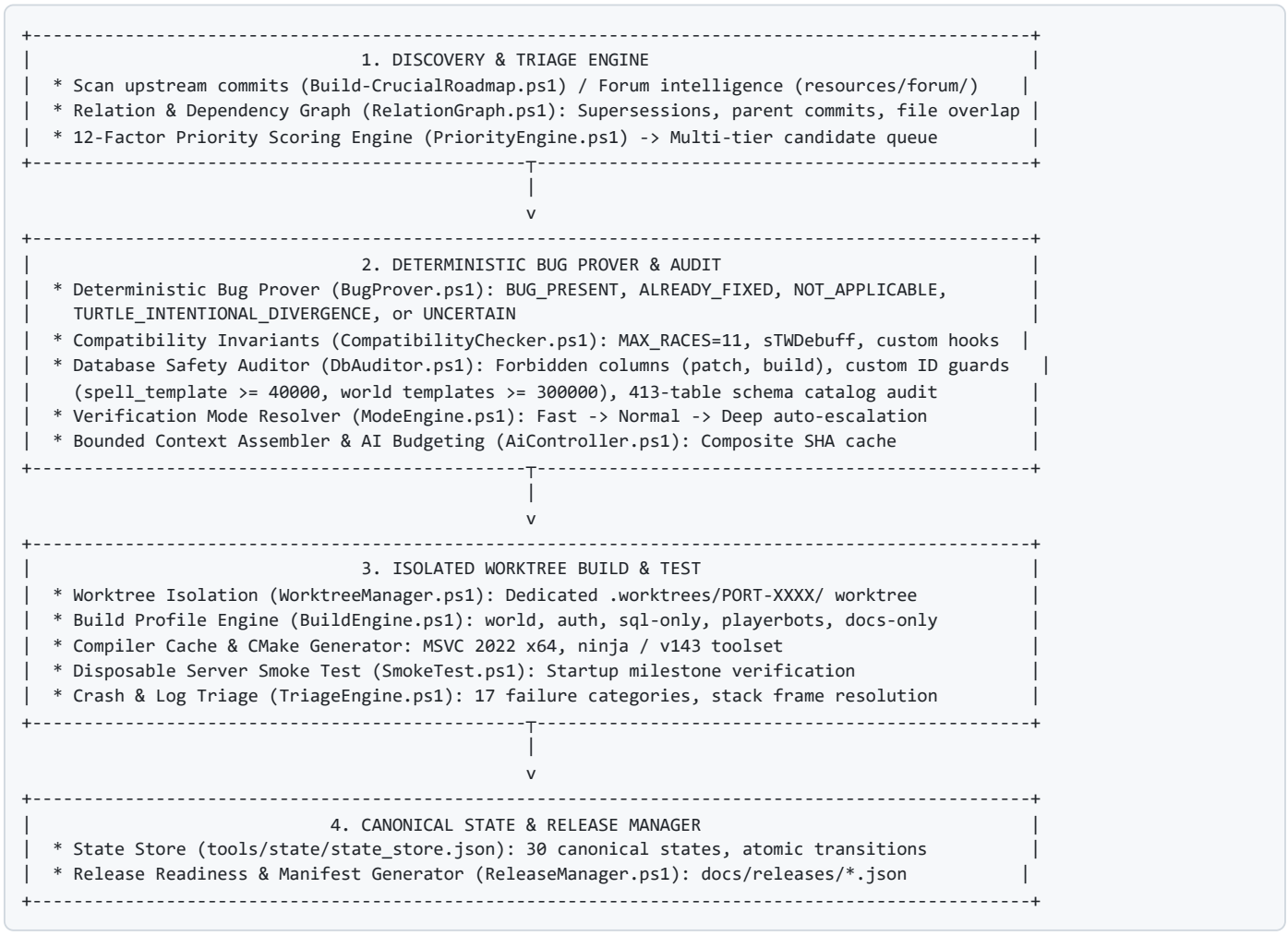
2. Architecture & Pipeline Flow

The orchestration architecture consists of three interconnected subsystems feeding through canonical state stores:



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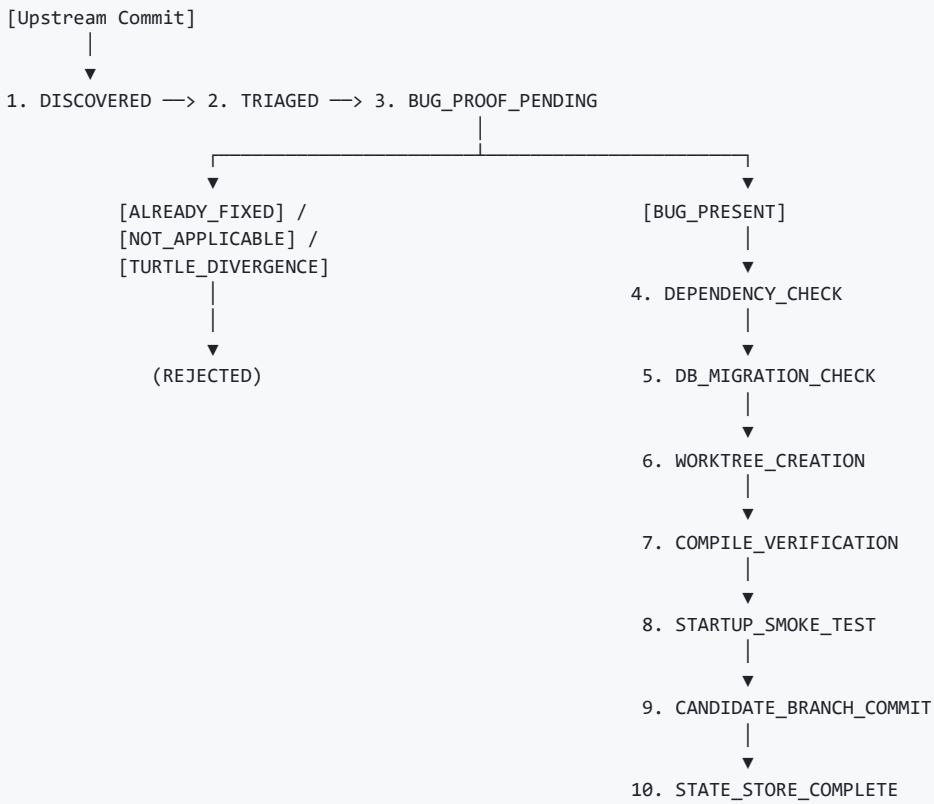
3. Workflow Lifecycle of a Candidate Fix



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4. Master Command Reference: Preserved Existing Commands

Command	Full Syntax	Mode	Access	AI Usage	Build Usage	DB Usage	Description
`task 1`	`task.ps1 1`	Normal	Write (Worktree)	None	Full MSVC	Offline Audit	Compiles, verifies, and packages all staged `PORT-XXXX` and `CORE-XXXX` packages.
`task 2`	`task.ps1 2 <sha/topic>`	Fast	Read-only	None	None	None	Searches 22,155 indexed forum threads for bug discussions and mechanics lore.
`task 3`	`task.ps1 3 [tbl] [id]`	Fast	Read-only	None	None	Offline Catalog	Audits pending database migration SQL or scalps table entity details.
`task 4`	`task.ps1 4 <sha>`	Normal	Read-only	Advisory	None	None	Generates bounded AI semantic dossier with touched code and context lines.
`task 5`	`task.ps1 5 <topic>`	Normal	Read-only	Advisory	None	Offline Catalog	Native Turtle core specification auditor for missing custom features.
`task 6`	`task.ps1 6 <id/query>`	Fast	Read-only	None	None	Online DB API	Queries official Turtle Online DB viewer for item, spell, and creature tooltips.
`task scalp`	`task.ps1 scalp <tbl> <id> [-Diff] [-Export]`	Fast	Read-only	None	None	Brotalnia/Base	Extracts entity definitions and generates side-by-side vanilla vs Turtle diffs.
`task extract`	`task.ps1 extract <tbl> <id>`	Fast	Read-only	None	None	Brotalnia/Base	Alias for `task scalp`.
`task port`	`task.ps1 port <sha> [-Mode Fast|Deep] [-DryRun] [-AutoCommit]`	Normal|	Variable	Write (Worktree)	Advisory	Patch-Aware	Migration Audit	Unified porting pipeline for upstream donor commits.
`task port-batch`	`task.ps1 port-batch <N> [-Tier 1-5] [-Mode M] [-DryRun] [-AutoCommit]`	Variable	Write (Worktree)	Advisory	Patch-Aware	Migration Audit	Batch executes porting pipeline on next N curated candidate commits.
`task auto-port`	`task.ps1 auto-port <sha> [-Mode M] [-DryRun]`	Variable	Write (Worktree)	Advisory	Full MSVC	Migration Audit	One-command end-to-end port & commit for single commit (Tasks 2->3->4->1).
`task auto-pilot`	`task.ps1 auto-pilot [N] [-Tier 1-5] [-Mode M]`	Variable	Write (Worktree)	Advisory	Full MSVC	Migration Audit	One-command autonomous batch port & commit for N commits (Tasks 2->3->4->1).
`task restore`	`task.ps1 restore <topic> [-StageTemplate]`	Normal	Staging	Advisory	None	Offline Catalog	Audits official staff posts and stages native core restoration manifests.
`task restore-batch`	`task.ps1 restore-batch <N>`	Normal	Staging	Advisory	None	Offline Catalog	Batch audits next N un-audited Turtle patch topics from roadmap queue.

Command	Full Syntax	Mode	Access	AI Usage	Build Usage	DB Usage	Description
`task status`	`task.ps1 status`	Fast	Read-only	None	None	None	Displays live system metrics, queue counts, HEAD SHAs, and active run IDs.
`task pdf`	`task.ps1 pdf`	Fast	Read-only	None	None	None	Compiles `COMMAND_REFERENCE.md` to HTML and exports high-quality PDF via Edge.

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5. Master Command Reference: New Extended Commands

Command	Full Syntax	Mode	Access	AI Usage	Build Usage	DB Usage	Description
`task test`	`task.ps1 test`	Fast	Read-only	None	None	None	Executes comprehensive 38-case Pester test suite covering all invariants.
`task prove`	`task.ps1 prove <sha>`	Fast	Read-only	None	None	None	Deterministic bug prover inspecting diff hunks and target AST existence.
`task relations`	`task.ps1 relations <sha>`	Fast	Read-only	None	None	None	Analyzes commit reverts, duplicate fixes, and file overlap relationships.
`task dependencies`	`task.ps1 dependencies <sha>`	Fast	Read-only	None	None	None	Inspects git commit DAG for missing parents and prerequisite commits.
`task rank`	`task.ps1 rank`	Fast	Read-only	None	None	None	Computes 12-factor priority scores (0.0–100.0) across all pending candidates.
`task next`	`task.ps1 next`	Fast	Read-only	None	None	None	Returns the single highest-priority eligible candidate commit to port.
`task plan`	`task.ps1 plan <sha> [-Mode M]`	Variable	Read-only	None	None	None	Generates a complete verification plan and dry-run report without disk writes.
`task build-profile`	`task.ps1 build-profile <sha>`	Fast	Read-only	None	None	None	Selects optimal build target (`world` , `auth` , `sql-only` , `playerbots`).
`task smoke`	`task.ps1 smoke [-Mode M]`	Fast/Deep	Execute	None	Binary Test	None	Disposable server startup smoke test evaluating `mangosd.exe` & `realmd.exe` .
`task crash`	`task.ps1 crash <path>`	Fast	Read-only	None	None	None	Parses server crash dump / log, categorizes failure, and extracts stack frames.
`task triage`	`task.ps1 triage <logfile>`	Fast	Read-only	None	None	None	Categorizes log errors across 17 structured failure categories.
`task baseline`	`task.ps1 baseline [-Force]`	Fast	Read-only	None	Binary Test	None	Checks and caches compile, link, and startup health of target repository HEAD.
`task state`	`task.ps1 state`	Fast	Read-only	None	None	None	Outputs canonical pipeline state store summary and active run statistics.
`task config-check`	`task.ps1 config-check`	Fast	Read-only	None	None	None	Validates repository paths, toolchain binaries, and schema configuration.
`task state-check`	`task.ps1 state-check`	Fast	Read-only	None	None	None	Verifies state store integrity, active runs, and candidate transition history.
`task worktree`	`task.ps1 worktree`	Fast	Read-only	None	None	None	Lists all currently active and managed isolated Git worktrees.
`task worktree-cleanup`	`task.ps1 worktree-cleanup [-DryRun]`	Fast	Write (Worktree)	None	None	None	Safely purges obsolete or completed worktrees inside `.worktrees/` .
`task parity`	`task.ps1 parity`	Fast	Read-only	None	None	Offline DBC	Audits binary client DBCs (`ChrRaces` , `Map` , `Spell`) against server code.
`task release-check`	`task.ps1 release-check`	Fast	Read-only	None	None	Offline Catalog	Evaluates all release readiness gates (tree clean, invariants pass, baseline).
`task release-manifest`	`task.ps1 release-manifest [name]`	Fast	Write (Docs)	None	None	None	Generates a signed machine-readable release manifest in `docs/releases/` .
`task tag-release`	`task.ps1 tag-release <name>`	Fast	Write (Git Tag)	None	None	None	Gated Git tag creation; rejects release if any readiness gates fail.
`task roadmap-refresh`	`task.ps1 roadmap-refresh [-FetchLatest]`	Fast	Read/Write (Roadmap)	None	None	None	Audits all 7,300+ upstream commits, refreshes queue, and fetches latest commits.

6. Verification Modes: FAST vs NORMAL vs DEEP

The system provides three strictly defined verification modes. Each mode enforces all safety invariants, but varies in compilation scope, runtime validation depth, and AI assistance:

Verification Mode	Code Analysis	Invariant Scan	DB Schema Audit	Build Execution	Startup Smoke Test	Runtime Test	AI Consultation	Typical Duration
Fast	Deterministic diff matching	Full manifest invariant scan	Full 413-table column audit	Skipped	Skipped	Skipped	Skipped (Deterministic only)	1 – 3 seconds
Normal	Diff + Nearby context AST	Full manifest invariant scan	Full 413-table column audit	Patch-Aware (`world` / `auth`)	Disposable startup check	Optional / Smoke	Advisory (On ambiguity only)	30 – 90 seconds
Deep	Full call-graph & packet trace	Full manifest invariant scan	Full 413-table column audit	Full MSVC Solution	Full disposable smoke test	Reproducer verification	Advisory + Contextual trace	2 – 5 minutes

Mandatory Common Checks (Enforced Across ALL Modes):

1. **Deterministic Bug-Existence Proof**: Verification that the defect exists in target and was not previously fixed or intentionally diverged.

2. **Compatibility Invariant Scan**: Verifies `MAX_RACES = 11`, `sTWDebuff`, `SCRIPT_COMMAND_TAKE_MONEY = 93`, and protected manager hooks.

3. **Database Migration Safety Check**: Forbidden progressive versioning columns (`patch`, `build`) and custom ID boundaries (`spell_template` >= 40000, world templates >= 300000).

4. **Target Working Tree Cleanliness**: Main checkout is verified clean before worktree creation.

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7. Automatic Mode Escalation Rules

The mode engine (`ModeEngine.ps1`) evaluates candidate risk factors. It automatically escalates execution modes to protect server integrity:

Fast $\rightarrow$ Normal Escalation Triggers:

- Candidate introduces or modifies SQL database migration files (`sql/database_updates/`).
- Candidate possesses one or more unresolved predecessor commit dependencies.
- Bug-existence confidence score is below 0.85.
- Candidate modifies core spell mechanics (`SpellMgr.cpp`, `SpellEffects.cpp`).

Normal $\rightarrow$ Deep Escalation Triggers:

- Commit subject or diff touches security keywords: `crash`, `packet`, `opcode`, `thread`, `mutex`, `deadlock`, `leak`, `exploit`, `auth`, `crypto`.
- Touched files reside in security or networking subsystems (`src/shared/Auth/`, `src/game/WorldSocket.cpp`, `src/game/Opcodes.cpp`).
- Concurrency constructs are modified (locks, atomic variables, threading queues).

> **IMPORTANT**
> **No Automatic Downgrade Policy** Under no circumstances will an explicitly requested mode be automatically downgraded (e.g., `-Mode Deep` will never execute as `Normal` or `Fast`).

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8. DryRun Mode & Safety Guarantees

Running any porting or cleanup command with `-DryRun` guarantees that **zero modifications** are written to the target repository or filesystem:

```
# Plan candidate 448df9ba0 without touching repository
task.ps1 port 448df9ba0 -DryRun
```

DryRun Guarantees:

- No Git branches created (`port/PORT-XXXX` is not created).
- No Git worktrees spawned (`.worktrees/` remains unchanged).
- No compiler or linker invocations executed.
- No database files modified.
- Outputs complete execution plan: Bug verdict, priority score, resolved mode, required build profile, and dependency links.

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9. Auto-Pilot Mode & Autonomous Batch Processing

Auto-Pilot Mode enables hands-free, continuous candidate evaluation, bug proving, worktree provisioning, patch normalization, compatibility auditing, and candidate staging across multiple commits without manual per-commit intervention.

The One-Command End-to-End Pipeline: Tasks 2 -> 3 -> 4 -> 1

When you want an autonomous, zero-friction pipeline that takes upstream commits all the way to candidate branch commits in a single pass, use **Auto-Pilot Mode**:

```
# Auto-Pilot a batch of 10 candidates with automatic compilation and git commit:
task.ps1 auto-pilot 10

# Auto-Pilot a specific single commit:
task.ps1 auto-port 448df9ba0

# Or via the -AutoCommit switch:
task.ps1 port-batch 10 -AutoCommit
task.ps1 port 448df9ba0 -AutoCommit
```

How the Chained Pipeline Works:

1. **Task 2: Forum & Mechanics Intelligence Scout (`Search-ForumArchive.ps1`)**:

Automatically mines 22,155 indexed Turtle-WoW forum threads using keywords extracted from the commit subject. Identifies any related mechanics discussions, player bug reports, or staff statements.

2. **Task 3: Database & Migration Safety Audit (`Audit-DatabaseMigrations.ps1`)**:

Checks whether the upstream commit touches SQL migrations. Inspects table definitions against the 413-table schema catalog, verifies Turtle custom ID ranges (creature >= 300,000, spell >= 40,000), checks for forbidden progressive columns, and stages any required SQL files into `tools/queue/staging_sql/` .

3. **Task 4: AI Context Assembly & Semantic Dossier (`Invoke-AiAudit.ps1`)**:

Performs bounded diff context extraction, checks surrounding code ASTs, evaluates Turtle custom divergences, verifies hard invariants (`MAX_RACES=11` , `sTWDebuff`), generates an AI audit dossier in `tools/queue/ai_dossiers/<sha>.md` , and packages the candidate into `tools/queue/02_ready_to_build/PORT-XXXX.json` .

4. **Task 1: Isolated Worktree Builder & Committer (`Build-ReadyPackages.ps1`)**:

Creates an isolated git worktree (`.worktrees/candidate-PORT-XXXX`), applies the patch safely, enforces build profiles (`world` , `auth` , `sql-only`), compiles via MSVC 2022, and commits to a candidate branch with full provenance recorded in `tools/state/state_store.json` .

FAQ: If I run `task.ps1 port-batch 10` , do I have to run `task.ps1 1` after?

Execution Command	Staged to Queue?	Automatically Compiled?	Automatically Committed?	Need to run `task 1` after?
`task.ps1 port-batch 10`	Yes (`02_ready_to_build/`)	No	No	**YES** (Review first, then compile via `task 1`)
`task.ps1 auto-pilot 10`	Yes	Yes (in worktree)	Yes (candidate branch)	**NO** (Fully automated in 1 command)
`task.ps1 port-batch 10 -AutoCommit`	Yes	Yes (in worktree)	Yes (candidate branch)	**NO** (Fully automated in 1 command)
`task.ps1 auto-port <sha>`	Yes	Yes (in worktree)	Yes (candidate branch)	**NO** (Fully automated in 1 command)

> **NOTE**
> - Use `task.ps1 port-batch 10` when you want a **review step** (inspecting AI dossiers and patches in `tools/queue/02_ready_to_build/` before compiling). - Use `task.ps1 auto-pilot 10` or `-AutoCommit` when you want a **hands-off single command** that finishes the entire process and commits verified candidate branches automatically.

Clean CLI Invocations (From Default Open Location / Any Directory)

You do **not** need to `cd` into the project repository. All modules dynamically resolve repository roots from `$PSScriptRoot`.

1. Direct Invocation from Windows Command Prompt (`cmd.exe`):

```
powershell.exe -ExecutionPolicy Bypass -File "C:\Users\Admin\AntigravityProfiles\Projects\twow project\tools\task.ps1" autc
```

2. Direct Invocation from PowerShell (from `C:\Users\Admin>` or any path):

```
& "C:\Users\Admin\AntigravityProfiles\Projects\twow project\tools\task.ps1" auto-pilot 10
```

3. Permanent Global Setup via PowerShell `$PROFILE` (Recommended):

Run this once in PowerShell to make `task` globally accessible from any terminal:

```
if (!(Test-Path $PROFILE)) { New-Item -ItemType File -Path $PROFILE -Force | Out-Null }
Add-Content $PROFILE "`nfunction task { & 'C:\Users\Admin\AntigravityProfiles\Projects\twow project\tools\task.ps1' @args }
```

Once added, open a new PowerShell prompt anywhere and simply type:

```
task auto-pilot 10
task status
task next
task roadmap-refresh
```

Running Auto-Pilot Mode: With Tier vs Without Tier

A. Running WITH Tier Specification:

When you want to focus exclusively on a specific severity or architectural domain:

```
# Port the next 10 Tier 1 candidates (Crashes, Leaks, Deadlocks, Exploits)
task.ps1 port-batch 10 -Tier 1 -Mode Normal

# Port the next 10 Tier 2 candidates (Combat Accuracy, Spells, Formulas)
task.ps1 port-batch 10 -Tier 2 -Mode Normal
```

B. Running WITHOUT Tier Specification:

When you omit the `-Tier` argument:

```
# Auto-Pilot without tier: ports the next 10 highest-value candidates
task.ps1 port-batch 10 -Mode Normal
```

How Does Auto-Pilot Choose Commits Without a Tier?

1. **Natural Architectural Hierarchy in `CRUCIAL_COMMITS_QUEUE.csv`**:

The master queue is pre-sorted by architectural urgency: **Tier 1 (Crashes/Exploits)** $\rightarrow$ **Tier 2 (Combat/Spells)** $\rightarrow$ **Tier 3 (NPC/AI)** $\rightarrow$ **Tier 4 (Movement/Maps)** $\rightarrow$ **Tier 5 (General)**. Without a tier filter, Auto-Pilot processes strictly down this natural hierarchy.

2. **Dynamic 12-Factor Priority Scoring Engine (`task rank` / `task next`)**:

In addition to tier order, every candidate is scored dynamically from 0.0 to 100.0 by `PriorityEngine.ps1`:

- Crash Severity: $+25$ points
- High Player Impact: $+20$ points
- Security / Exploit Risk: $+20$ points
- Low Target File Churn: $+15$ points
- Dependency Satisfaction: $+10$ points
- Baseline Stability: $+10$ points
- Non-Applicable / Intentional Divergence: **0.0 score hard gate** (instantly skipped).

You can inspect this ranking at any time with `task rank` or retrieve the single best candidate with `task next`.

3. **State Store & Git History Deduplication**:

Auto-Pilot automatically cross-references `tools/state/state_store.json` and the target `tortoise-wow` git history. Any candidate that has already been evaluated (`ALREADY_FIXED`, `NOT_APPLICABLE`, `TURTLE_INTENTIONAL_DIVERGENCE`, `PACKAGE_READY`, or `RELEASED`) is **automatically bypassed**, ensuring zero redundant work.

The Recommended 3-Phase Auto-Pilot Runbook

To run Auto-Pilot with 100% safety and predictability:

```
# Phase 1: Pre-Flight DryRun (Evaluates bug prover and outputs plan without writing files)
task.ps1 port-batch 10 -DryRun

# Phase 2: Isolated Autonomous Porting (Runs in .worktrees/PORT-XXXX/ with safety checks)
task.ps1 port-batch 10 -Mode Normal

# Phase 3: Single-Writer Candidate Build & Branch Commit
task.ps1 1
```

Auto-Pilot Safety Guarantees:

- **Zero Working Tree Contamination**: All patch application and test compilation occur strictly in `.worktrees/PORT-XXXX/`. The main repository working tree remains 100% untouched.
- **Stale Package Auto-Invalidation**: If the target repository HEAD advances during a batch run, any staged package whose pinned base SHA does not match is automatically invalidated.
- **Non-Destructive Rollback**: If a patch fails compilation or invariant checks, the isolated worktree is safely deleted without ever executing `git reset --hard` or `git checkout .`.

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10. Roadmap Research, Commit Auditing & Catching New Upstream Commits (`task roadmap-refresh`)

The upstream VMaNGOS repository (`vmangos/core`) continuously accepts new bugfixes and accuracy patches. The orchestration framework includes an automated research and auditing engine to catch, classify, and queue these new commits:

```
# Audit all local upstream commits and regenerate the roadmap
task.ps1 roadmap-refresh

# Fetch new commits from GitHub upstream remote, merge development, and audit
task.ps1 roadmap-refresh -FetchLatest
```

How the Roadmap List Refreshes:

1. **Upstream Remote Fetch (`-FetchLatest`)**: When `-FetchLatest` is passed, the engine runs `git fetch origin` and `git merge --ff-only origin/development` inside `reference-upstreams/vmangos-core`, pulling down all new upstream commits.

2. **Donor Commit Scanning**: Scans all 7,300+ upstream commits in repository history (newest to oldest).

3. **Cross-Referencing Port History**: Compares every upstream commit SHA against commits already merged into `tortoise-wow` and documented in `docs/BACKPORT_HISTORY.md`.

4. **Older Superseded Duplicate Elimination**: Detects intermediate or superseded fixes and removes older duplicates so only modern final fixes are queued.

5. **Architectural Tier Classification**: Automatically assigns each commit to Tier 1 through Tier 5 based on modified subsystems and commit subjects.

6. **Artifact Generation**:

- Updates `tools/porting/CRUCIAL_COMMITS_QUEUE.csv` (the active queue of 5,092 deduplicated crucial candidates).
- Updates `tools/porting/ALL_AVAILABLE_COMMITS_REFERENCE.csv` (complete 7,339-commit catalog).
- Re-renders `docs/ROADMAP.md` with updated metrics and live backlog status.

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11. Build Profiles & Compiler Toolchain

The build engine (`BuildEngine.ps1`) optimizes build times by targeting only the solution components touched by the candidate patch:

Build Profile	Affected Subsystems	MSVC Target Project	Typical Compile Time
`world`	`src/game/` , `src/scripts/`	`mangosd`	~45 seconds
`auth`	`src/realmd/` , `src/shared/Auth/`	`realmd`	~15 seconds
`sql-only`	`sql/` , documentation	None (Bypasses compilation)	Instant (0s)
`playerbots`	`src/game/playerbot/`	`mangosd` (PlayerBots profile)	~45 seconds
`docs-only`	`docs/` , `tools/`	None	Instant (0s)

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12. Baseline Health Check & SHA Pinning

The baseline checker (`BaselineChecker.ps1`) verifies the compile, link, and startup state of the unchanged target repository before applying candidate patches:

```
task.ps1 baseline
```

- **Cached Baseline**: Once verified for a target HEAD SHA, baseline status is cached in `tools/state/state_store.json`.
- **Three-Tier Classification**:
- `COMPILE_FAIL` : Target source fails compilation prior to patch.
- `LINK_FAIL` : Compilation passes, but linker errors occur.
- `STARTUP_FAIL` : Binary links, but crashes on startup (e.g. missing DBC or DB config).
- `HEALTHY` : Compile, link, and startup all succeed.

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13. Bug-Existence Proving Engine

The bug prover (`BugProver.ps1`) executes deterministic analysis before any candidate is ported:

```
task.ps1 prove 448df9ba0
```

Deterministic Verdicts:

- `BUG_PRESENT`: Target source file contains the exact unpatched code pattern or cleanly applicable bug signature.
- `ALREADY_FIXED`: Target source already contains the upstream fix or equivalent logic from previous commits.
- `NOT_APPLICABLE`: Commit modifies files or subsystems that do not exist in Tortoise-WoW.
- `TURTLE_INTENTIONAL_DIVERGENCE`: Target code intentionally diverges from vanilla (e.g., custom race handling, debuff engine).
- `UNCERTAIN`: Context has diverged significantly; requires manual or advisory AI review.

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14. Database Safety, Schema Catalog & Provenance

Database migrations undergo rigorous automated static analysis against `config/schema_catalog.json` (413 verified tables):

```
task.ps1 3 "sql/database_updates/world/20260507165648_world.sql"
```

Enforced Rules:

1. **Forbidden Progressive Columns**: Rejects any statement containing `patch`, `build`, `min_patch`, or `max_patch`.

2. **Custom Spell Boundary Protection**: `spell_template` IDs ≥ 40000 are reserved for Turtle-WoW custom spells. Modifications to these IDs trigger a validation failure.

3. **Custom World Entity Protection**: `creature_template`, `item_template`, `quest_template`, and `gameobject_template` IDs ≥ 300000 are reserved for Turtle custom content.

4. **Statement Anchoring**: Parser anchors table matching (`(?m)^\s` *) to prevent false positives from quest text strings like* "update my count"*.

5. **Entity Provenance Tracking**: Records source donor revision, target entity ID, original value, proposed value, and confidence rating in structured dossiers.

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15. DBC, Client & Core Parity Auditor

The parity auditor (`ParityAuditor.ps1`) parses binary WDBC client data from `reference-upstreams/client-data-1.18.1/dbc` and verifies server constants:

```
task.ps1 parity
```

- **ChrRaces.dbc**: Verifies playable race records against `#define MAX_RACES 11` in `SharedDefines.h`.
- **Map.dbc**: Audits 57 map definitions against server map enum declarations.
- **Spell.dbc**: Verifies custom spell entry boundary rules.
- **ItemClass.dbc**: Verifies item classification bitmasks.

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16. Crash Dump & Server Log Triage

Automated triage engine (`TriageEngine.ps1`) categorizes runtime issues and crash dumps across 17 distinct failure categories:

```
# Triage server startup or runtime log
task.ps1 triage "tortoise-wow/bin/Release/Server.log"

# Triage crash dump or assertion trace
task.ps1 crash "tortoise-wow/bin/Release/crash.dmp"
```

Supported Failure Classifications:

`ASSERTION_FAILURE`, `SEGMENTATION_FAULT`, `DATABASE_ERROR`, `DBC_ERROR`, `OPCODE_ERROR`, `MAP_LOAD_ERROR`, `SPELL_ERROR`, `SCRIPT_ERROR`, `NETWORK_ERROR`, `CONFIG_ERROR`, `AUTH_ERROR`, `MEMORY_LEAK`, `DEADLOCK`, `STARTUP_TIMEOUT`, `SHUTDOWN_HANG`, `CUSTOM_SYSTEM_ERROR`, `UNKNOWN_FAILURE`.

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17. Worktree Isolation & Cleanup Management

All candidate modifications are strictly isolated to Git worktrees:

```
# View all managed active worktrees
task.ps1 worktree

# Safe cleanup of merged or obsolete worktrees
task.ps1 worktree-cleanup
```

Isolation Rules:

- Located exclusively in `<TargetRepo>/.worktrees/PORT-XXXX/`.
- Rooted on ephemeral branch `port/PORT-XXXX-<sha>`.
- Pruned cleanly using `git worktree remove` upon pipeline completion or rejection.
- Safety check prevents deletion of any directory outside `.worktrees/`.

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18. Run IDs, Idempotency & Resumption

Every pipeline execution generates a unique, sortable Run ID:

```
RUN-yyyyMMdd-HHmmss-xxxx  (e.g., RUN-20260909-144022-7a1b)
```

- **Idempotent Resumption**: State store (`tools/state/state_store.json`) tracks active runs and stage transitions.
- Interrupted runs resume from the last certified state without re-running deterministic proofs.

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19. Canonical 30-State Machine

Candidates transition through a strictly guarded 30-state lifecycle:

State	Category	Description	Allowed Next States
`DISCOVERED`	Initial	Candidate identified from upstream donor commit	`TRIAGED`, `BUG_PROOF_PENDING`, `ALREADY_FIXED`, `NOT_APPLICABLE`, `REJECTED`
`TRIAGED`	Priority	Categorized and scored by Priority Engine	`BUG_PROOF_PENDING`, `ALREADY_FIXED`, `NOT_APPLICABLE`, `TURTLE_INTENTIONAL_DIVERGENCE`, `REJECTED`
`BUG_PROOF_PENDING`	Proving	Undergoing deterministic diff & AST analysis	`BUG_PRESENT`, `ALREADY_FIXED`, `NOT_APPLICABLE`, `TURTLE_INTENTIONAL_DIVERGENCE`, `UNCERTAIN`
`BUG_PRESENT`	Verified Defect	Bug proved present in target source	`DEPENDENCY_PENDING`, `DEPENDENCY_READY`, `DB_CHECK_PENDING`, `ADAPTATION_REQUIRED`, `PATCH_READY`
`ALREADY_FIXED`	Terminal Clean	Upstream fix is already present in target	`NEEDS_REAUDIT`
`NOT_APPLICABLE`	Terminal Clean	System/file not present in target architecture	`NEEDS_REAUDIT`
`TURTLE_INTENTIONAL_DIVERGENCE`	Guarded	Target intentionally behaves differently	`HUMAN_REVIEW_REQUIRED`, `REJECTED`, `NEEDS_REAUDIT`
`UNCERTAIN`	Ambiguous	Cannot prove deterministically	`HUMAN_REVIEW_REQUIRED`, `ADAPTATION_REQUIRED`, `REJECTED`
`DEPENDENCY_PENDING`	Dependency	Waiting for predecessor commit to be ported	`DEPENDENCY_READY`, `HUMAN_REVIEW_REQUIRED`, `REJECTED`
`DEPENDENCY_READY`	Dependency	All required predecessor commits satisfied	`DB_CHECK_PENDING`, `ADAPTATION_REQUIRED`, `PATCH_READY`
`DB_CHECK_PENDING`	Database	Undergoing schema catalog and column audit	`DB_CHECK_PASS`, `DB_CHECK_FAIL`
`DB_CHECK_PASS`	Database	Schema valid, no forbidden columns, no collisions	`ADAPTATION_REQUIRED`, `PATCH_READY`
`DB_CHECK_FAIL`	Database	Invariant violation or column collision detected	`ADAPTATION_REQUIRED`, `HUMAN_REVIEW_REQUIRED`, `REJECTED`
`ADAPTATION_REQUIRED`	Adaptation	Requires code adjustment for Turtle custom systems	`PATCH_READY`, `HUMAN_REVIEW_REQUIRED`, `REJECTED`
`PATCH_READY`	Staged	Patch cleanly formatted and ready for isolated build	`COMPILE_PENDING`, `REJECTED`
`COMPILE_PENDING`	Build	Worktree created; queued for compilation	`COMPILE_PASS`, `COMPILE_FAIL`, `BLOCKED_BY_BASELINE`
`COMPILE_PASS`	Build	MSVC compilation succeeded with 0 errors	`STARTUP_PENDING`, `RUNTIME_PENDING`, `COMPLETE`
`COMPILE_FAIL`	Build	MSVC compilation failed	`BLOCKED_BY_BASELINE`, `ADAPTATION_REQUIRED`, `HUMAN_REVIEW_REQUIRED`, `REJECTED`
`STARTUP_PENDING`	Smoke Test	Binary launched in disposable environment	`STARTUP_PASS`, `STARTUP_FAIL`, `BLOCKED_BY_BASELINE`
`STARTUP_PASS`	Smoke Test	Binary initialized cleanly without crashes	`RUNTIME_PENDING`, `COMPLETE`, `HUMAN_REVIEW_REQUIRED`
`STARTUP_FAIL`	Smoke Test	Binary crashed or asserted on startup	`BLOCKED_BY_BASELINE`, `ADAPTATION_REQUIRED`, `REJECTED`
`RUNTIME_PENDING`	Runtime	Queued for in-game reproducer verification	`RUNTIME_PASS`, `RUNTIME_FAIL`
`RUNTIME_PASS`	Runtime	In-game behavior matches vanilla specification	`COMPLETE`, `HUMAN_REVIEW_REQUIRED`
`RUNTIME_FAIL`	Runtime	In-game reproducer failed	`ADAPTATION_REQUIRED`, `HUMAN_REVIEW_REQUIRED`, `REJECTED`
`HUMAN_REVIEW_REQUIRED`	Approval Gate	Ambiguous fix flagged for human decision	`HUMAN_APPROVED`, `HUMAN_REJECTED`
`HUMAN_APPROVED`	Approval Gate	Human operator certified fix for inclusion	`PATCH_READY`, `COMPILE_PENDING`, `COMPLETE`

State	Category	Description	Allowed Next States
`HUMAN_REJECTED`	Approval Gate	Human operator rejected fix	`REJECTED`
`BLOCKED_BY_BASELINE`	Baseline Block	Blocked because unchanged target HEAD is broken	`NEEDS_REAUDIT` , `REJECTED`
`NEEDS_REAUDIT`	Invalidation	Invalidated due to target HEAD moving or staleness	`BUG_PROOF_PENDING` , `TRIAGED` , `DISCOVERED`
`COMPLETE`	Final Pass	Certified, tested, committed to candidate branch	`NEEDS_REAUDIT`
`REJECTED`	Final Fail	Rejected candidate permanently archived	`NEEDS_REAUDIT`

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20. CI / GitHub Actions PR Workflows

Two GitHub Actions workflows automate continuous integration across orchestration tools and candidate server builds:

1. `.github/workflows/orchestration-ci.yml` :
   - Runs on every push and pull request touching `tools/` , `config/` , or `docs/` .
   - Executes `task test` (Pester 38-case test suite).
   - Validates JSON schema contracts (`config/schemas/`).
   - Runs configuration health check (`task config-check`).
   - Verifies command reference and PDF generation.
2. `.github/workflows/server-candidate-ci.yml` :
   - Triggered when candidate port branches (`port/*`) are pushed.
   - Sets up MSVC 2022 toolchain and Ninja build system.
   - Runs baseline validation against target base commit.
   - Executes patch-aware compilation profile.
   - Uploads build logs and triage reports as artifacts.

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21. Release Checkpoints & Manifest Generation

Release commands provide certification gates before tagging or publishing server releases:

```
# Evaluate release readiness gates
task.ps1 release-check

# Generate certified release manifest
task.ps1 release-manifest "Release-1.18.1-Update1"

# Create signed git tag (aborts if readiness gates fail)
task.ps1 tag-release "v1.18.1-update1"
```

- Manifests are saved to `docs/releases/<name>.json` with full commit SHAs, toolchain info, and test certifications.

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22. Critical Safety Warnings

> **CAUTION**
> 1. **Single-Writer Constraint**: Never invoke `-AutoBuild` or compiler tasks concurrently in multiple shells. 2. **Never Commit Directly to Extended Main**: Candidate fixes must be isolated to candidate branches (`port/PORT-XXXX-<sha>`). 3. **Never Touch Working Tree**: Never execute `git checkout .` , `git reset --hard` , or `git clean -fd` in `tortoise-wow` . 4. **No Secrets in Logs**: Never print or commit API keys, authentication tokens, or private user passwords.

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23. Troubleshooting & Common Failure States

Error Code / Symptom	Root Cause	Solution
`EXIT CODE 1: MAX_RACES invariant violation`	Candidate patch alters `MAX_RACES` or loop bound.	Restore `#define MAX_RACES 11` in patch; adapt race array allocations.
`EXIT CODE 1: Custom ID boundary collision`	Patch uses `spell_template` entry ≥ 40000 or world entry ≥ 300000.	Renumber entity to vanilla range (< 40000 or < 300000) or scalp correct ID.
`EXIT CODE 1: Target repository has uncommitted changes`	Working tree is dirty; worktree isolation safety triggered.	Commit or stash changes in `tortoise-wow` before running porting commands.
`EXIT CODE 2: Microsoft Edge not found`	Edge binary missing at standard 32/64-bit location.	Verify installation of Edge or update path in `Export-CommandReferencePdf.ps1`.
`EXIT CODE 2: CMake executable not found`	`cmake.exe` not detected in PATH or vcpkg tools directory.	Run `task config-check` to verify auto-discovery or update `config/twow-project.json`.
`STALE PACKAGE: Target base SHA mismatch`	Target repo HEAD advanced since package was staged.	Re-audit candidate via `task port <sha>` against new target HEAD.

24. Expected Status & Verdict Reference Values

Category	Canonical Allowed Values	Meaning
Pipeline Status	`PASS`, `FAIL`, `WARN`, `SKIPPED`, `BLOCKED`	Stage execution result code
Bug Verdict	`BUG_PRESENT`, `ALREADY_FIXED`, `NOT_APPLICABLE`, `TURTLE_INTENTIONAL_DIVERGENCE`, `UNCERTAIN`	Bug prover determination
Verification Mode	`Fast`, `Normal`, `Deep`	Verification intensity level
Build Profile	`world`, `auth`, `sql-only`, `playerbots`, `docs-only`	Selected MSVC build target
AI Usage	`NONE`, `ADVISORY_ONLY`, `AMBIGUITY_SYNTHESIS`	AI role in candidate evaluation
Exit Codes	`0` (PASS), `1` (VALIDATION_FAIL), `2` (TOOL_FAIL), `3` (SCHEMA_FAIL)	Process return codes

25. Architectural Comparison: Why the 2.0 Build System is Vastly Superior to the Legacy 1.0 System

The 2.0 Total Revamp overhaul transforms the repository from a collection of fragile manual porting scripts into a high-assurance, non-destructive, enterprise-grade autonomous engineering framework:

Architectural Dimension	Legacy 1.0 Porting System	Revamp 2.0 Autonomous Architecture	Tangible Engineering Advantage
Git Working Tree Safety	Directly modified target repository working tree; polluted checkout.	Strict Git worktree isolation in ephemeral `.worktrees/PORT-XXXX/`.	Target repo checkout is **100% clean and immune** to accidental corruption or debris.
Rollback & Cleanup	Ran destructive `git checkout .`, `git reset --hard`, and `git clean -fd`.	Non-destructive: simply unlinks the worktree (`Remove-IsolatedWorktree`).	Eliminates catastrophic data loss risk; never discards developer uncommitted work.
Commit Staging & Branches	Committed directly to active target branch or left loose patches in folders.	Changes committed exclusively to candidate branches (`port/PORT-XXXX-<sha>`).	Enables clean PR reviews, branch audits, and CI smoke testing before merge.
Bug Existence Verification	Blindly applied patches; failed on Turtle custom code divergences.	Deterministic Bug Prover (`BugProver.ps1` / `task prove`).	Classifies `BUG_PRESENT`, `ALREADY_FIXED`, `NOT_APPLICABLE`, or `TURTLE_DIVERGENCE` before touching code.
Build Speed & Disk Footprint	Recompiled full solution (`world` + `auth`) for every patch (~15-30 min per commit).	Patch-Aware Build Profiles (`world`, `auth`, `sql-only`, `docs-only`).	Bypasses compilation for SQL/docs (0s), targets single daemons (15-45s), saving hours of build time.
Database Migration Safety	Unverified SQL execution; potential collision with custom Turtle content.	Cached 413-table schema catalog (`DbAuditor.ps1`) and strict boundary guards.	Blocks forbidden progressive columns (`patch`, `build`) and protects custom ranges (`spell` $\ge 40k$, world $\ge 300k$).
Client Data Parity	Manual inspection or reliance on external assumptions.	Automated binary WDBC parser (`ParityAuditor.ps1` / `task parity`).	Guarantees code parity with 1.18.1 client DBCs (`MAX_RACES = 11`, maps, spell definitions).
State Tracking & Resumption	Disparate queue directories with loose JSON files prone to desync.	Atomic canonical JSON state store (`state_store.json`) with 30-state transition guards.	Deterministic run IDs (`RUN-yyyyMMdd-HHmmss-xxxx`), transition guards, and resume capability.
AI Token Efficiency	Unbounded prompts; risked spending tokens on already-fixed candidates.	Deterministic-first gating, bounded context (≤ 200 lines), composite SHA256 caching.	0 AI tokens spent on deterministic rejections; prevents hallucination via `ADVISORY_ONLY`.
Runtime Reliability	No startup validation; crashes only discovered after manual server launch.	Disposable startup smoke tests (`SmokeTest.ps1`) and 17-category log triage.	Catches assertion failures, heap corruptions, and missing DBCs before candidate commits are certified.
Automated Testing & CI	Zero automated tests; scripts were untested in continuous integration.	Comprehensive 38-spec Pester suite (`task test`) + 2 GitHub Actions CI workflows.	Sub-7-second automated verification ensuring every invariant, schema, and command passes.